Lab 7: Null Space and the Rank-Nullity Theorem Solutions

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, October 21st, 2026

Name: _____

uniqname: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Recap: Rank, Column Space, and Null Space

- For an $n \times d$ matrix A, $\text{colsp}(A)$ is a subspace of $\mathbb{R}^n$; it is the span of A's columns.
- $\text{rank}(A) = \dim(\text{colsp}(A))$.
- $\text{nullsp}(A) = \{\vec{x} \in \mathbb{R}^d : A\vec{x} = \vec{0}\}$.
- The **Rank-Nullity Theorem** states that

$$\text{rank}(A) + \dim(\text{nullsp}(A)) = d$$

- $\text{rank}(A^T) = \text{rank}(A)$.

Activity 1: Null Space of a Matrix with Linearly Independent Columns

Let $A = \begin{bmatrix} 3 & 0 \\ 0 & 4 \\ 1 & 0 \end{bmatrix}$. What is $\text{nullsp}(A)$?

Solution: The columns of A are linearly independent. So, the only way to combine them to get $\vec{0}$ is to use zero for both coefficients. This means

$$\boxed{\text{nullsp}(A) = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}}$$

Notice that the null space contains the zero vector in $\mathbb{R}^2$, not $\mathbb{R}^3$, since A has two columns.

Activity 2: Null Space from Column Relationships

Let $X = \begin{bmatrix} 1 & 2 & -1 & 3 & 4 & 4 \\ 2 & 5 & -2 & 7 & 11 & 10 \\ 4 & 8 & -4 & 12 & 16 & 16 \end{bmatrix}$.

Note that X's first two columns form a basis for its column space; verify this before proceeding.

a) Fill in the blanks: nullsp(X) is a ____-dimensional subspace of ____.

Solution: The Rank-Nullity Theorem tells us that

$$2 + \dim(\text{nullsp}(X)) = 6$$

so $\dim(\text{nullsp}(X)) = 4$. Since X has six columns, the vectors in its null space have six components.
So, the blanks are $\boxed{4}$ and $\boxed{\mathbb{R}^6}$.

b) Find a basis for nullsp(X). *Hint: You should be able to answer this without solving equations.*

Solution: Let $\vec{v}_j$ be column j of X. From the columns, we can see that

$$\vec{v}_3 = -\vec{v}_1, \qquad \vec{v}_4 = \vec{v}_1 + \vec{v}_2, \qquad \vec{v}_5 = -2\vec{v}_1 + 3\vec{v}_2, \qquad \vec{v}_6 = 2\vec{v}_2$$

Moving everything to one side in each equation gives a linear combination of the columns equal to $\vec{0}$. The coefficients give us four vectors in nullsp(X):

$$\boxed{\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ 0 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \\ 0 \\ -1 \end{bmatrix} \right\}}$$

These vectors are linearly independent: in components 3 through 6, each vector has a non-zero entry where the other three have zeros. Since the null space has dimension 4, these four vectors form a basis.

Activity 3: Four Spaces Warm-Up

Let A be a 7×9 matrix with rank 5. For each space below, state its dimension and the space $\mathbb{R}^k$ of which it is a subspace.

Space	Dimension	Subspace of $\mathbb{R}^k$ (state k)
colsp(A)		
colsp(A^T)		
nullsp(A)		
nullsp(A^T)		

Solution: The completed table is:

Space	Dimension	Subspace of
colsp(A)	5	$\mathbb{R}^7$
colsp(A^T)	5	$\mathbb{R}^9$
nullsp(A)	4	$\mathbb{R}^9$
nullsp(A^T)	2	$\mathbb{R}^7$

Both column spaces have dimension 5, since rank$(A) =$ rank$(A^T) = 5$. For the null spaces, apply the Rank-Nullity Theorem to A and A^T:

$$\dim(\text{nullsp}(A)) = 9 - 5 = 4, \qquad \dim(\text{nullsp}(A^T)) = 7 - 5 = 2$$

Remember that A^T has shape 9×7.

Activity 4: Constructing a Matrix

Find a matrix A such that

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} \in \text{colsp}(A) \qquad \text{and} \qquad \begin{bmatrix} 1 \\ 3 \end{bmatrix} \in \text{nullsp}(A).$$

Solution: The first condition tells us that A needs two rows, and the second tells us that it needs two columns.

Let's make the first column $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Then the first condition is automatically satisfied. For the second condition, we need the first column plus three times the second column to be $\vec{0}$. So, one answer is

$$\boxed{A = \begin{bmatrix} 3 & -1 \\ 1 & -1/3 \end{bmatrix}}$$

We can check that

$$A \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3-3 \\ 1-1 \end{bmatrix} = \vec{0}$$

There are many other correct answers!

Activity 5: The Systems of Equations View

Let A be an $n \times d$ matrix of rank r, and suppose there exist vectors $\vec{b} \in \mathbb{R}^n$ such that

$$A\vec{x} = \vec{b}$$

does not have a solution (meaning no $\vec{x}$ makes $A\vec{x} = \vec{b}$).

a) What are all inequalities ($<$ or $\leq$) that must be true between n, d, and r?

Solution: If $A\vec{x} = \vec{b}$ has no solution, then $\vec{b}$ is not in colsp(A). So, the columns of A don't span all of $\mathbb{R}^n$, which means $r < n$.
Also, A only has d columns, so it can't have more than d linearly independent columns. This gives

$$\boxed{r < n \qquad \text{and} \qquad r \leq d}$$

We can't conclude whether $n < d$, $n = d$, or $n > d$; all three are possible.

b) How do you know that $A^T\vec{y} = \vec{0}$ has solutions other than $\vec{y} = \vec{0}$?

Solution: From part (a), $r < n$. Since A^T has n columns and rank r, the Rank-Nullity Theorem gives

$$\dim(\text{nullsp}(A^T)) = n - r > 0$$

So, nullsp(A^T) contains more than just the zero vector. Any non-zero vector in this null space is a solution to $A^T\vec{y} = \vec{0}$.

Activity 6: Columns in a Null Space

Let A and B be non-zero 6×6 matrices. Suppose $\text{rank}(A) = 4$, and that every column of B is in $\text{nullsp}(A)$.

a) Show that every row of A is orthogonal to every column of B.

Solution: Take any column $\vec{b}$ of B. Since $\vec{b} \in \text{nullsp}(A)$, we know that $A\vec{b} = \vec{0}$.
Each entry of $A\vec{b}$ is the dot product of a row of A with $\vec{b}$. Since every entry is zero, every row of A is orthogonal to $\vec{b}$. This works for every column of B.

b) Find AB and $\text{rank}(AB)$. Justify your answers.

Solution: Each column of AB is A times the corresponding column of B. All of these products are $\vec{0}$, so

$$\boxed{AB = 0_{6\times 6} \qquad \text{and} \qquad \text{rank}(AB) = 0}$$

The zero matrix has no linearly independent columns.

c) What are the possible values of $\text{rank}(B)$? Explain why these are the only possibilities, and why each one is possible.

Solution: By the Rank-Nullity Theorem, $\dim(\text{nullsp}(A)) = 6-4 = 2$. Every column of B is in this two-dimensional space, so at most two columns of B can be linearly independent. Also, B is non-zero, so its rank can't be 0.
This leaves $\boxed{\text{rank}(B) \in \{1, 2\}}$. Both are possible. Let $\{\vec{u}, \vec{v}\}$ be a basis for $\text{nullsp}(A)$:

- For rank 1, make every column of B equal to $\vec{u}$.
- For rank 2, make the first two columns $\vec{u}$ and $\vec{v}$, and make the remaining columns $\vec{0}$.

Both choices satisfy all of the conditions in the question.

Activity 7: Can the Column Space Equal the Null Space?

In each part, determine whether it's possible to find a matrix whose column space and null space are the same. If it is, give an example and show that it works. If not, explain why not.

a) Can you find a 2×2 matrix A such that $\text{colsp}(A) = \text{nullsp}(A)$?

Solution: Yes! One example is

$$\boxed{A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}$$

Its column space is span $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$. To find its null space, note that

$$A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ 0 \end{bmatrix} = \vec{0} \quad \iff \quad x_2 = 0$$

So, its null space is also span $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$. Both spaces are the horizontal axis in $\mathbb{R}^2$.

b) Can you find a 3×3 matrix A such that $\text{colsp}(A) = \text{nullsp}(A)$?

Solution: No. If the column space and null space were equal, they would have the same dimension. Calling this dimension r, the Rank-Nullity Theorem would give

$$r + r = 3 \quad \implies \quad r = \frac{3}{2}$$

But the dimension of a subspace must be an integer, so this is impossible.